2.

**Given**
- Avg size of macadamia nuts ~ 20 mm
- Avg max grip strength ~ 50.7 kg
- Avg load to crack nut ~ 222.18 kg

**Find**
- Design to make this feasible
- Discuss usability
- Modify to use linear actuator

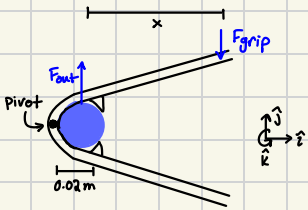
**Plan**
- Convert units
- Make diagram
- Find length
- Find MA
- Compare actuators

**(a)**

20 mm = 0.02 m

50.7 kg = 497.20 N

222.18 kg = 2178.84 N




$\Sigma M_p = 0$

$F_{out}(0.02) - F_{grip}(x) = 0$

$497.2x = 0.02(2178.84)$

$x = 0.088\ m$

looking at when $F_{out} = F_{crack}$

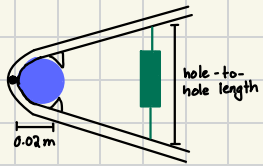
$MA = \frac{F_{out}}{F_{in}} = \frac{2178.84}{497.2} = 4.38$

**(b)** This design is useable because the mechanical advantage of 4.38 allows enough to crack the nut with a normal grip strength. However, the handle length of 0.088 m is fairly short, which could make it harder to hold comfortably. Overall, the design should work in theory, but a longer handle would probably make it easier to use

**(c)**

Looking for an actuator that can produce a force ≥ 2178.84 N
↓
489.82 lbf

I chose model PA-03. The force of this model is 600 lbs which is enough force to crack the nut with a reasonable margin.




for stroke length of 4"

Actuator starts at extended length which is 14.88". It then pulls the handles together as it moves to the retracted length which is 10.88".

This is not very usable since the handles would have to be very far apart for the actuator to fit.